Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

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University of Melbourne

MAST20030: Differential Equations
Tutorial Solutions: Week 9

Solution to Question 1.

(a) The general solution to the ODE is $y(x) = C_1 e^{-x} + C_2 x e^{-x}$. Applying the first condition gives $y(0) = C_1 = 1$. We then have $y'(x) = -e^{-x} + C_2 e^{-x} - C_2 x e^{-x}$ so that $y'(2) = -e^{-2} + C_2 e^{-2} - 2C_2 e^{-2} = (-1 - C_2)e^{-2} = 0$ therefore $C_2 = -1$. The solution to the BVP is therefore $y(x) = (1 - x)e^{-x}$.

(b) The general solution to the ODE and its derivative are as given in (a). Applying the first condition gives $y(0) = C_1 = 0$. The second condition gives $y'(1) = -e^{-1} + C_2 e^{-1} - C_2 e^{-1} = -e^{-1} \neq 1$ and so the BVP has no solution.

(c) The general solution to the ODE is $y(x) = x + C_1 \cos x + C_2 \sin x$ and so $y'(x) = 1 - C_1 \sin x + C_2 \cos x$. The first condition gives $y'(0) = 1 + C_2 = 1$ and so $C_2 = 0$. Applying the second condition gives $y'(\pi) = 1 - C_2 = 1$ which is consistent with the first condition in that $C_2 = 0$. C_1 is therefore arbitrary and so the BVP has an infinite number of solutions given by $y(x) = x + C_1 \cos x$.

Solution to Question 2.

The eigenvalue equation for $-\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ is

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2} f(x) = \lambda f(x) \;\Rightarrow\; f''(x) + \lambda f(x) = 0. \tag{1}$$

If $\lambda = 0$, the solutions have the form $f(x) = Ax + B$, where $A, B \in \mathbb{R}$ (or $\mathbb{C}$). The boundary conditions require that $0 = f'(0) = A$ and $0 = f(\pi) = A\pi + B$, i.e., $A = B = 0$. Thus, 0 is not an eigenvalue.
If $\lambda \neq 0$, then $f(x) = C_1 \cos x\sqrt{\lambda} + C_2 \sin x\sqrt{\lambda}$. Applying $f'(0) = 0$ immediately gives $C_2 = 0$. Applying $f(\pi) = 0$ requires $\cos \pi\sqrt{\lambda} = 0$ which gives

$$\lambda_n = \left(n - \frac{1}{2}\right)^2, \qquad n = 1, 2, 3, \ldots.$$

The corresponding eigenfunctions are

$$f_n(x) = \cos\left(n - \frac{1}{2}\right)x, \qquad n = 1, 2, 3, \ldots$$

Solution to Question 3.

(a) If we let $\phi(x, t) = X(x)T(t)$, then $X(x)T'(t) = 4X''(x)T(t)$. Rearranging gives $\dfrac{T'(t)}{4T(t)} = \dfrac{X''(x)}{X(x)}$ and so the only way for this to be true for all x and t is if both sides are equal to a constant. If we choose this constant to be $-\lambda$ then we arise at an eigenvalue problem for X, i.e., $-X''(x) = \lambda X(x)$. Substituting our separable solution into the boundary conditions gives $X'(0)T(t) = 0$ and $X(\pi)T(t) = 0$ and so provided the solution is not trivial, i.e., $T(t) \neq 0$, then we have $X'(x) = 0$ and $X(\pi) = 0$ which is exactly our eigenvalue problem from Question 2.

(b) Since we have the eigenvalues, we can solve our T equation,

$$\frac{T_n'(t)}{4T_n(t)} = -\left(n - \frac{1}{2}\right)^2, \qquad n = 1, 2, 3, \ldots$$

which has solution

$$T_n(t) = C_n \mathrm{e}^{-4\left(n-\frac{1}{2}\right)^2 t}$$

We therefore have a general solution which satisfies the PDE and boundary conditions given by

$$\phi(x,t) = \sum_{n=1}^{\infty} C_n \mathrm{e}^{-4\left(n-\frac{1}{2}\right)^2 t} \cos\left(n - \frac{1}{2}\right) x$$

(c) Applying the initial condition gives

$$\phi(x,0) = \sum_{n=1}^{\infty} C_n \cos\left(n - \frac{1}{2}\right) x = \cos\frac{5x}{2}.$$

From matching coefficients we can see that $C_3 = 1$ and all other C_n must be zero (we could also prove this using orthogonality). The complete solution is therefore

$$\phi(x,t) = \mathrm{e}^{-25t} \cos\frac{5x}{2}$$

(d) Calculating the partial derivatives and substituting the initial and boundary conditions confirms this solution satisfies the IBVP.

Solution to Question 4.

(a) With the new boundary conditions, $\lambda = 0$ results in $B = A$ and $A\pi + B = A$, hence $A\pi = 0$ so that 0 is not an eigenvalue. For $\lambda \neq 0$, we again have $f(x) = C_1 \cos x\sqrt{\lambda} + C_2 \sin x\sqrt{\lambda}$. Applying the two conditions gives

$$C_1 = \sqrt{\lambda} C_2 \qquad \text{and} \qquad C_1 \cos\pi\sqrt{\lambda} + C_2 \sin\pi\sqrt{\lambda} = \sqrt{\lambda}\left(-C_1 \sin\pi\sqrt{\lambda} + C_2 \cos\pi\sqrt{\lambda}\right)$$

which on rearranging leads to

$$(1+\lambda)\sin\pi\sqrt{\lambda} = 0.$$

We therefore have the eigenvalues $\lambda_n = n^2$ for $n = 1, 2, 3, \ldots$ and $\lambda_{-1} = -1$, and hence $\sqrt{\lambda_{-1}} = i$. This gives the following eigenfunctions

$$f_n(x) = \sin nx + n\cos nx \qquad \text{and} \qquad f_{-1}(x) = -i(i\cos ix + \sin ix) = \mathrm{e}^x$$

where we have chosen $C_2 = -i$ so that $f_{-1}(x)$ is real for real arguments.

(b) Separation of variables leads to the eigenvalue problem in (a). The separated solutions to the T equations are $T_n(t) = C_n \mathrm{e}^{-4n^2 t}$ and $T_{-1}(t) = C_{-1}\mathrm{e}^{4t}$. The general solution is therefore

$$\phi(x,t) = C_{-1}\mathrm{e}^{4t}\mathrm{e}^x + \sum_{n=1}^{\infty} C_n \mathrm{e}^{-4n^2 t}\left(\sin nx + n\cos nx\right).$$

To fix the constants given an initial condition, we would need to derive the orthogonality properties of the eigenfunctions (try this yourself if you're brave).